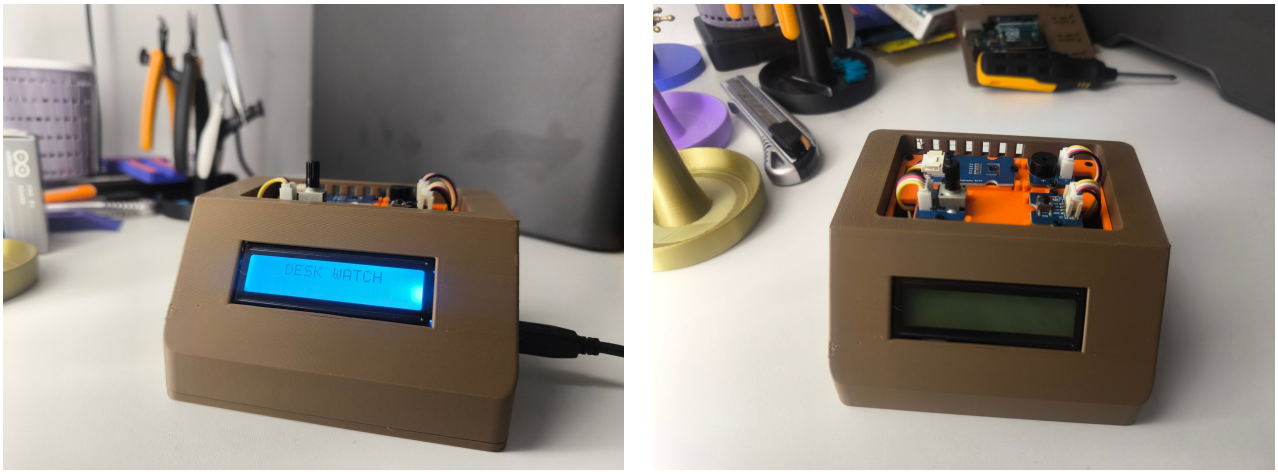
Desk Watch 1

User guide and build reference – MELODY and TEACHER editions

Arduino UNO + Grove desktop console | Pr. BELAMINE MakerLab



Printed prototype: start-up screen (left), and the open tray with the Grove modules (right).

Desk Watch 1 is a compact desktop console built around an Arduino UNO and a Grove Base Shield. It shows temperature and humidity on a 16x2 colour LCD, warns you when the room leaves the range you want, and also works as a timer, stopwatch, counter, alarm clock, dice, chess clock and melody player. Everything is controlled with **one button and one dial**, and every module plugs in with a Grove cable: no soldering. The **TEACHER** edition adds an **EXAM** application: exam countdown and paper collection on a single screen.

At a glance

- **Climate station**: SHT31 or SHT40 sensor (detected automatically), adjustable HIGH / LOW thresholds, alerts by colour and by buzzer. Every sensor reading is checked (CRC-8); a corrupted reading is ignored.
- **Tools**: Pomodoro timer, counter, stopwatch, alarm clock (eight ringtones), dice (D2 to D100), two-player chess clock, melody player (four tunes).
- **RTC clock required** (DS1307 or DS3231): it provides the time, the alarm and the automatic night mode.
- **One colour per page**: the backlight changes with the page, and the menu takes the colour of the page it points at.
- Two editions, two languages each: **MELODY** (10 menu entries) and **TEACHER** (11 entries, with EXAM), in an FR or EN version. The language is set by the version, not by a setting.
- Fits an Arduino UNO: about 82 % of the program memory for MELODY and 88 to 89 % for TEACHER, about a third of the RAM, with no library other than the Grove display one.

1. What you need

Part	Notes
Arduino UNO R3	Any UNO R3 compatible board
Grove Base Shield for Arduino	Keep the voltage switch on 5 V
Grove LCD RGB Backlight V4.0 (16x2)	I2C. Uses the *Grove - LCD RGB Backlight* library (rgb_lcd.h)
Grove temperature and humidity sensor SHT31 (or SHT40)	I2C, address 0x44. The type is detected automatically
Grove Buzzer	Digital port D3
Grove Button	Digital port D4
Grove Rotary Angle Sensor (the dial)	Analog port A1
Grove RTC (DS1307 or DS3231)	I2C, address 0x68. **Required**: time, alarm and automatic night mode
Grove cables, USB-B cable, small screws	See the assembly guide for the screw plan

2. Wiring

Plug each Grove cable into the matching port of the Base Shield. The three I2C devices (display, humidity sensor and RTC) can share any I2C ports: they are in parallel.

Module	Base Shield port	Signal
LCD RGB display	I2C	SDA / SCL
SHT31 / SHT40 sensor	I2C	SDA / SCL (address 0x44)
RTC	I2C	SDA / SCL (address 0x68)
Buzzer	D3	Digital output, square wave (pulse-width modulation for the volume)
Button	D4	Digital input, HIGH when pressed
Dial (Rotary Angle Sensor)	A1	Analog input, 0 to 1023

The small L LED on the Arduino board (pin 13) lights up in sleep mode. Its colour depends on the board (often orange). For a red indicator, plug a red Grove LED into a free port and change `PIN_LED` at the top of the sketch.

3. Installing the firmware

Step	What to do
1	Install the Arduino IDE (2.x), then, in the library manager, install **Grove - LCD RGB Backlight** (Seeed Studio). No other library is needed: the humidity sensor and the RTC are read directly over I2C.
2	Pick the edition and the language: `Desk_Watch_1_MELODY_Edition_EN.ino`, `_MELODY_Edition_FR.ino`, `Desk_Watch_1_TEACHER_Edition_EN.ino` or `_TEACHER_Edition_FR.ino`. Keep each file in the folder of the same name, open it, choose **Arduino UNO** and your port, then upload.
3	At start-up you see DESK WATCH, a short melody if the start-up sound is on, then PR BELAMINE MAKERLAB, the edition loading bar and a module check (RTC and sensor).
4	If the dial turns the wrong way with your wiring, or sits on another port, adapt `potRead()` and `PIN_POT` at the top of the sketch. If the shield switch is on 3.3 V, set `POT_MAX` to 675.

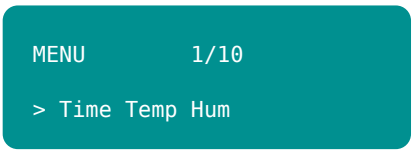
4. Controls

The Desk Watch has one button and one dial. The same gestures work on every page.

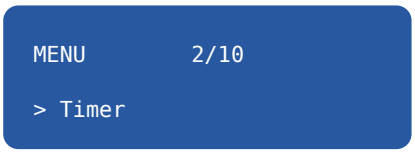
Gesture	Effect
Short press	Opens the menu from the home page; picks the entry in the menu; confirms a setting; main action of a page. It is confirmed 0.3 s after release, to leave room for a double click.
Double click	Two presses within 0.3 s. Page-specific action: reset the timer or stopwatch, set the thresholds on Temp & Hum, switch the alarm off, pause the chess clock, back to the menu from several pages.
Long press (0.8 s or more)	Back: opens the menu from a page, cancels a setting, goes back one step. The timers and the stopwatch keep running in the background. During an EXAM session (TEACHER) it opens its commands.
Dial	Its position picks the menu entry or the value of a setting: turn right for the next entry or a larger value. In the menu, the dial only takes over once turned: the menu opens on the entry of the page you are leaving.
Sleep	**Sleep mode** menu entry: the screen goes off, the L LED lights up, the alarm stays active. A press or a turn of the dial wakes the device. There is no sleep gesture: no very long press.

The menu

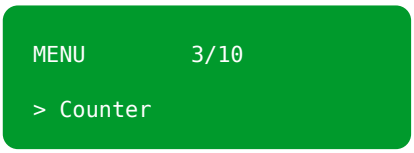
Each menu entry takes the colour of its page.



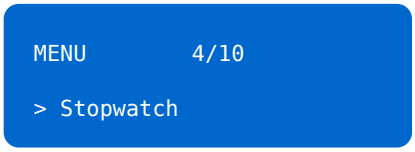
1/10



2/10



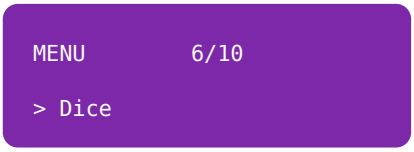
3/10



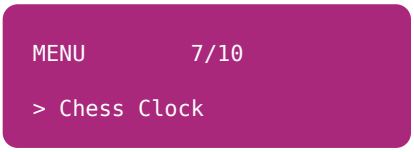
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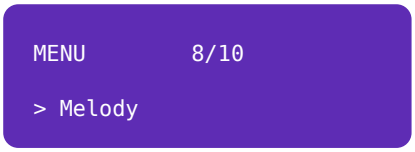
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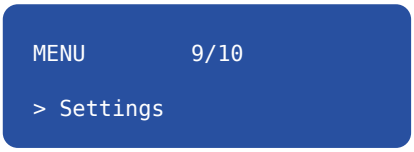
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9/10



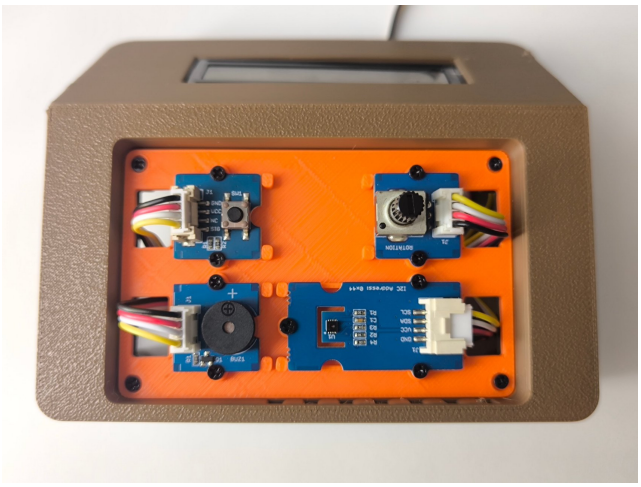
10/10

Entry	What it opens
Exam (TEACHER only)	First entry of TEACHER: exam countdown and paper collection
Time Temp Hum	Clock, temperature, humidity and alerts
Timer	Pomodoro timer, from 1 min to 4 h
Counter	Hand counter
Stopwatch	Stopwatch
Alarm	Set and enable the alarm clock
Dice	Dice with 2 to 100 sides
Chess Clock	Two-player chess clock
Melody	Melody player: four tunes
Settings	Brightness, volume, start-up sound, auto sleep, night mode, time, favorite, factory reset
Sleep mode	Turns the screen off; the alarm stays active

The **favorite** is not a menu entry: it is chosen in Settings and sets the page shown at start-up.

Module layout

Tray seen from above, the screen being at the top of the picture. Each module is held by screws and plugs into the Base Shield with a short Grove cable that goes down through the side openings. The photographed prototype tray carries an SHT31; the SHT40 mounts in the same place.



Top left the button, top right the dial, bottom left the buzzer, bottom right the humidity sensor.

Top-right indicators

When a function runs in the background, a letter appears in the top-right corner of the screen: **!** temperature or humidity alert, **A** alarm enabled, **C** stopwatch running, **M** timer running (the M goes off for 1 s every 10 s).

Backlight colours

Each page has its own colour, so you know where you are at a glance. The station and timer colours also change with their state.

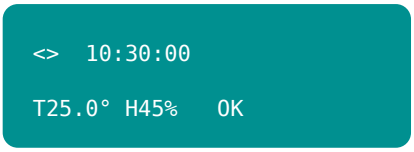
Page	Colour	When
Temp & Hum		In range (turquoise)
		Too high (red)
		Too low (blue)
Timer		Setting the time (blue)
		Work (orange)
		Paused (yellow)
		Short break (green)
		Long break (cyan)
Counter		
Stopwatch		
Alarm		
Dice		
Chess clock		Other player's turn: orange
Melody		
Settings		
Sleep mode (menu)		Red, then the screen goes off
EXAM		Running (dark green); time is up: red; all papers in: green

5. The pages

Temp & Hum

The station home page. Line 1 shows the RTC time and the background-function indicators; line 2 shows temperature and humidity. The colours give the status at a glance: **turquoise** in range, **red** too high, **blue** too low. When a threshold is crossed, line 2 alternates every 2 s between the readings and the alert message, the clock staying visible. The buzzer signals the alert (silent in night mode and in sleep). A small hysteresis (0.3 C and 2 %) stops the alert from flickering around a threshold.

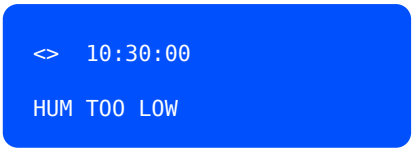
- **Dial**: changes the display (<> arrows left of the clock): readings, minimum, maximum, average, then "AUT" which cycles through the four displays every 3 s.
- **Double click**: set the four thresholds (Tmin, Tmax, Hmin, Hmax). The dial picks the value, a short press goes to the next threshold then saves, a long press goes back to the previous threshold. Ranges: temperature 10 to 40 C in 0.5 C steps, humidity 10 to 90 %.
- Default values: 20.0 to 28.0 C and 40 to 60 % humidity. Thresholds are saved and reused at start-up.
- **Short press**: opens the menu, on the entry of the page.
- If the sensor is unplugged or returns corrupted data three times in a row, the screen shows "Sensor missing".



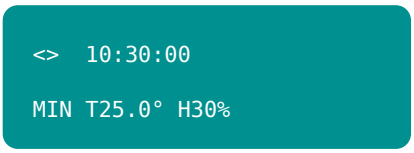
In range



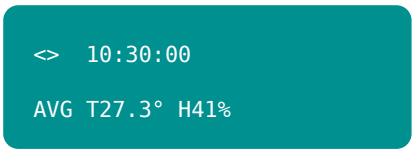
Temperature alert



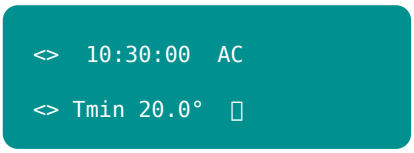
Humidity alert



Minimum



Average



Setting the thresholds

Timer (Pomodoro)

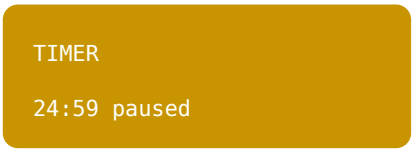
Turn the dial to choose the work time: 1 to 60 minutes in 1-minute steps, then 65 minutes to 4 hours in 5-minute steps. A short press starts or pauses; a double click resets (then, if already at zero, goes back to the menu). After each work period comes a 5-minute break, and a 15-minute break every 4th session. Once started, turning the dial opens the menu: the timer keeps running in the background. The colour shows the phase (see the colour table).



Choosing the time



Work



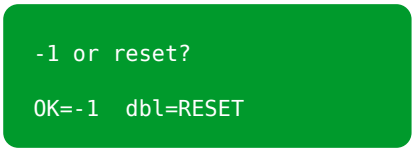
Paused

Counter and Stopwatch

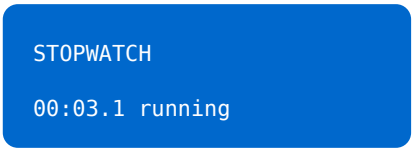
- **Counter**: each short press adds 1 immediately, even if you click fast. Turning the dial opens the action screen: short press = -1, double click = reset, 5 s without an answer or a long press = cancel.
- **Stopwatch**: a short press starts or stops, a double click resets. Already at zero, a double click goes back to the menu. Turning the dial opens the menu: the stopwatch keeps running in the background.



Counter



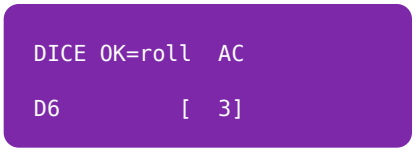
Turning the dial



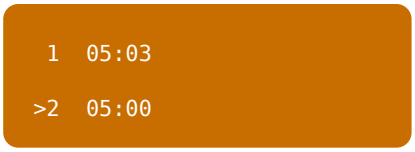
Stopwatch running

Dice, Chess clock and Melody

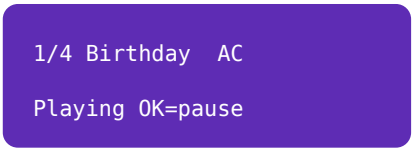
- **Dice**: the dial picks the number of sides (2, 4, 6, 8, 10, 12, 20, 30 or 100). A short press rolls: the digits scroll with a rattle for about a second, then the result stays on screen. A double click goes back to the menu.
- **Chess clock**: when stopped, the dial picks the time control (1+0, 3+0, 3+2, 5+0, 5+3, 10+0, 10+5, 15+10, 20+0, 30+0, 60+0: minutes + seconds added per move). A short press starts with player 1; after that each short press hands over to the other player. A double click pauses; then short press = resume, double click = reset, long press = menu. Auto sleep never triggers during a game.
- **Melody**: the dial picks a tune (Happy Birthday, Für Elise, Super Bros, Tetrix); a short press plays, pauses or resumes; a double click goes back to the menu.



Dice



Chess clock, player 2



Melody

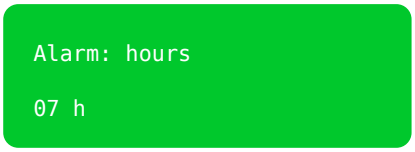
Alarm

Open **Alarm** in the menu. A short press enables and sets the alarm: hours, minutes, ringtone and volume (short press = next step, long press = previous step), the dial picks each value. The ringtone and the volume play a live preview. Eight ringtones are available: the four tunes of the player plus Aurora, Cosmic Pulse, Arcade Hero and Epic Rise. A double click disables the alarm; the letter A at the top right of the pages shows that it is enabled.

When it rings, any press or turn of the dial stops it for the day and opens the menu. With no reaction, it stops by itself after 120 s and rings again 5 minutes later, until you touch it. The alarm rings even in night mode and in sleep mode.



Alarm page



Hours

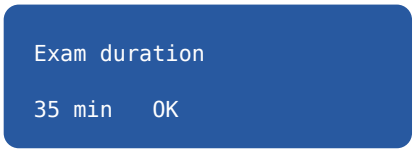


Ringtone (preview)

EXAM (TEACHER edition)

Open **Exam**, the first entry of the TEACHER menu. The dial picks the duration (1 to 240 minutes) then the expected papers (1 to 99), a short press confirms each step; the suggested values are 35 min and 24 papers. A last short press starts the session.

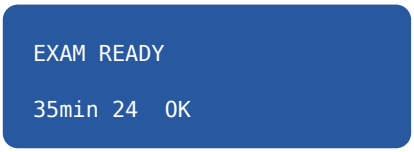
- Each short press counts **one paper** immediately, even after the time is up; two quick presses count two papers.
- A **long press** opens the commands: pick with the dial and confirm with a short press: back, subtract a paper, pause or resume, +5 min, end exam (with confirmation), main menu. A double click closes the commands.
- At zero, two short beeps sound and collecting stays possible. When all the papers are in, the screen turns green; the final message alternates with the actual working time, breaks excluded.
- The countdown keeps running in the background when you open the menu, and auto sleep is suspended during the session. The session stays in RAM: it does not survive a power cut.



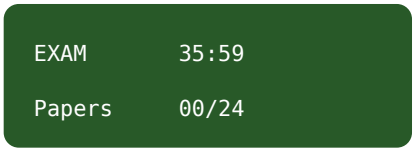
Duration



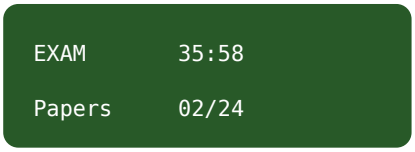
Expected papers



Ready



Collecting



Two papers in



Commands (long press)



Time is up



Alternates with the total

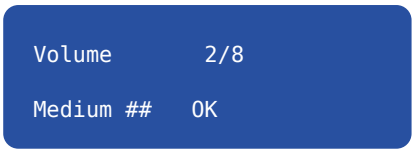
6. Settings

Open **Settings** in the menu. The dial picks a line, a short press opens it, the dial picks the value, a short press confirms. A long press cancels the change and goes back. Settings are saved in EEPROM.

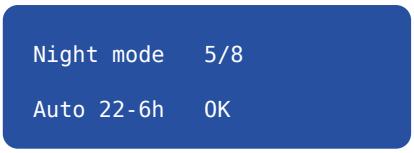
#	Setting	Choices	Notes
1	Brightness	8 levels, 5 to 100 %	Default 85 %. Applies to the backlight of every page; a long press cancels
2	Volume	Low / Medium / High	Changes the buzzer duty cycle; a test beep plays while you choose. Default: Medium
3	Startup sound	Yes / No	Four-note melody at start-up and when waking from sleep. Default: No
4	Auto sleep	Never, 1, 2, 5, 10, 15, 30 min, 1 h	Screen off after no press or dial movement. Default: Never
5	Night mode	No / Auto 22-6h / Yes	Screen reduced to 8 %, no alert beeps, no start-up melody. The alarm rings at full brightness. Auto uses the RTC. Default: Auto
6	Set time	Hours, then minutes	Writes the RTC; a long press cancels
7	Favorite	Menu, Time Temp Hum, Timer, Stopwatch, Counter, Dice, Chess clock (TEACHER: + Exam)	Page shown at start-up. "Menu": the menu opens at start-up. Default: Menu
8	Factory reset	One short press opens, a second confirms	Puts every setting, the thresholds and the alarm back to the original values; the RTC time is not touched



Brightness



Volume



Night mode

There is no language setting: each version (FR or EN) uses a single language.

Without the RTC

The RTC module is required. Without it the device starts and Temp & Hum stays usable (the time shows "--:--:--"), as do the timer, stopwatch, counter, dice, chess clock, melody player and EXAM. The clock, the time setting and the alarm then ask you to install the RTC module, and the automatic night mode is inactive.

7. Troubleshooting

Symptom	What to check
Backlight on but no text at start-up	Reinstall the Grove LCD library and upload again; check that the display cable is on an I2C port.
Sensor missing	Sensor cable on an I2C port; it must answer at address 0x44 (SHT31 or SHT40). The message also appears after three corrupted readings in a row.
"Install RTC module"	The RTC is required for the clock, the time setting and the alarm: plug it into an I2C port (address 0x68) and restart.
The dial turns backwards or jumps	Check that the dial is on A1 and the shield switch on 5 V (otherwise `POT_MAX` = 675). Reverse it in `potRead()` if needed.
The buzzer is too quiet or too loud	Settings, Volume (Low, Medium, High). Piezo buzzers differ; an active buzzer module ignores the duty cycle.
The screen is almost dark at night	Night mode is on (Settings 5). Set it to No.
The screen is off	The device is in sleep mode: a press or a turn of the dial wakes it up.
Compile error: not enough memory	MELODY uses about 82 % and TEACHER 88 to 89 % of the UNO program memory. Do not add large functions without removing something.

Desk Watch 1 is part of the Pr. BELAMINE MakerLab range, created by a teacher as a base for electronics projects in class. Your feedback and ideas are welcome in the comments of the model page, including requests for other languages and other editions.